

ADITYA ROY BARDHAN

Data Science & AI Student | AI/ML · Computer Vision · Software Engineering

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SUMMARY

Data Science & AI student at IIT Guwahati building practical AI systems, data-driven applications, and intelligent software. Experienced in computer vision, machine learning, distributed model training workflows, and authorized cybersecurity assessments. Passionate about solving real-world challenges with verified architectures and measurable impact.

EDUCATION

Indian Institute of Technology (IIT) Guwahati 2025 – 2029
Bachelor of Science (Hons.) in Data Science & Artificial Intelligence
Key Coursework: Machine Learning Theory, Statistical Inference, Deep Learning, Computer Vision, Natural Language Processing, Distributed Systems, Data Structures & Algorithms.

TECHNICAL SKILLS

Core Proficiency: Python, PyTorch, SQL, Pandas, NumPy, Computer Vision, Machine Learning, Data Structures
Working Knowledge: JavaScript, TypeScript, C++, TensorFlow, ONNX, Docker, Git & GitHub Actions, Linux (Ubuntu)
Exploring & Specializing: Large Language Models (LLMs), RAG, Agentic AI, Geospatial AI (NASA FIRMS), Distributed

PROFESSIONAL EXPERIENCE & ENGAGEMENTS

Independent ML Consultant / Contributor (Contract) — Distributed Training Dec 2025 – Feb 2026

- Contributed to large-scale distributed training workflows utilizing 3D parallelism (Data, Pipeline, and Tensor Parallelism via Megatron-LM and DeepSpeed ZeRO-3) across GPU cluster nodes.
- Curated and verified reasoning-trace datasets for knowledge distillation into compact student models, maintaining chain-of-thought coherence.
- Diagnosed loss spikes and gradient instabilities by conducting systematic data quality audits across preprocessing and batching pipelines.

Confidential Cybersecurity Engagements — Authorized Security Assessor Jan 2023 – Aug 2025

- Conducted authorized security assessments for financial-sector and enterprise clients under strict NDA, focusing on REST API security, threat modeling, and authentication controls.
- Identified and provided remediation roadmaps for 29 high/critical security vulnerabilities across multiple authorized client engagements.
- Designed rule-based and anomaly-based detection mechanisms for real-time transaction integrity monitoring and automated log alerts.

Technical Operations Contributor — Infrastructure & Reliability Jun 2021 – Aug 2024

- Maintained backend service health, low-latency synchronization monitoring, and uptime reporting for high-concurrency gaming server infrastructure.
- Collaborated on observability tooling, distributed telemetry log analysis, and rapid incident response.

KEY TECHNICAL PROJECTS

ASHA — Alerting Satellite Heat Analytics [\[Python, NASA FIRMS, Geospatial Data, FastAPI, Leaflet.js\]](#)

Engineered real-time satellite fire detection and thermal analytics covering 7 high-risk zones across India. Ingested MODIS/VIIRS feeds, implemented false-positive filtering algorithms, and delivered an interactive alerting interface. Reduced detection latency from hours to under 15 minutes with 94% validated classification accuracy.

Code: <https://github.com/aditya2909rb/ASHA>

OncoSML — Computational Cancer-Vaccine Learning Workflow [\[Python, cBioPortal API, PyTorch, GitHub Actions, MLOps\]](#)

Developed a self-maintaining machine learning pipeline for cancer genomic research. Automated real-time ingestion from cBioPortal and continuous literature updates from GitHub, with integrated validation checks and live monitoring dashboard for personalized vaccine candidate modeling.

Code: <https://github.com/aditya2909rb/sml-project>

Satellite Imagery for Global Change Detection [\[PyTorch, OpenCV, GDAL, NASA Earthdata, U-Net\]](#)

Built a pixel-level segmentation model detecting wildfire smoke plumes and landscape anomalies from geostationary satellite feeds. Built a custom semi-automated multispectral annotation pipeline; achieved 0.82 IoU on held-out test sets.

Code: <https://github.com/aditya2909rb>

Real-Time Demand Forecasting & Dynamic Pricing [\[XGBoost, LSTM, ONNX, Feature Store, Python\]](#)

Engineered high-frequency demand forecasting models for ride requests with 200+ engineered spatio-temporal features. Created automated data drift detection and dynamic surge pricing algorithms achieving sub-100ms inference latency via model quantization.

Code: <https://github.com/aditya2909rb>

Siddhant — Ancient Indian Yantra Reconstruction Engine [\[Three.js, SymPy, NetworkX, TypeScript, React\]](#)

Grand Finalist project at Smart India Hackathon 2025. Algorithmic engine that parses archaic Sanskrit geometric descriptions, resolves structural dependencies using graph algorithms, and renders interactive 3D mathematical models of historical astronomical instruments.

Code: <https://github.com/aditya2909rb>

Agentic AI Safety Evaluation [\[Local LLM, Raspberry Pi 5, Linux Cgroups, Python\]](#)

Empirical safety evaluation of an autonomous LLM agent with controlled local filesystem access. Analyzed multi-step execution behaviors under strict sandboxing, permission gates, and hardware resource boundaries.

Code: <https://github.com/aditya2909rb>

HONORS & ACHIEVEMENTS

Smart India Hackathon (SIH) 2025 — Grand Finalist

Nationwide competition organized by the Ministry of Education & AICTE. Selected as Grand Finalist for "Siddhant" among thousands of engineering teams across India.

GitHub Pro & Open Source Contributor

Maintain active repositories in machine learning, satellite data analysis, and systems security. Profile: github.com/aditya2909rb.

Competitive Problem Solving & Algorithmic Design

Active problem solver on LeetCode (leetcode.com/u/adityaroybardhan2909/) and Kaggle (kaggle.com/aditya2909rb). Interest in group theory & Rubik's cube optimization.

REFERENCES

Academic and professional references are available upon request.